Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Computer Science and Engineering Department

Auxiliary Examination

BE-COE/CSE Time: 3 Hours M. Marks: 100	**Course Code:** UCS310 **Course Name:** Database Management System **Name of Faculty:** Dr. Shashank Singh

Note: *Attempt any FIVE questions. All questions carry equal marks. Provide proper justification and neat diagrams wherever required. Assume suitable missing data, if any.*

Q. No	Questions	Marks
1	(a) Explain the different types of integrity constraints in DBMS. Discuss entity integrity and referential integrity constraints with suitable examples. (b) Discuss and differentiate between two-tier and three-tier DBMS architectures. Explain their advantages and limitations.	20
2	(a) Explain the Entity-Relationship (ER) model in detail. Discuss its components such as entities, attributes, relationships, cardinality constraints, and participation constraints. Illustrate your answer with suitable examples. (b) A hospital wants to computerize its patient management system. The following requirements have been gathered: • Each patient is identified by a unique Patient_ID, and has a name, age, and contact number. • Each doctor is identified by a unique Doctor_ID, has a name and specialization. • A doctor can attend to many patients, and a patient may be attended by many doctors over time. • Every visit of a patient to a doctor is recorded with a date and diagnosis. • Each patient may be prescribed multiple medicines during a visit, along with the dosage. Draw an E-R diagram for this system showing entities, attributes, keys, relationships, and cardinality/participation constraints. State any assumptions made.	20
3	Consider the following database schema for an e-commerce system. Formulate SQL statements for the given queries. Customer (CustomerID, Name, City) Product (ProductID, ProductName, Price, Category) Orders (OrderID, CustomerID, ProductID, Quantity, OrderDate) 1. Find the names of customers who have placed at least one order for a product in the 'Electronics' category. 2. Retrieve the list of customers who have never placed any order. 3. Find the names of products that have been ordered by customers from 'Delhi'. 4. List all products that have never been ordered. 5. Find the average price of products ordered by each customer.	20

<table>
<tr><th>Q. No</th><th>Questions</th><th>Marks</th></tr>
<tr><td>4</td><td>Consider the relation R (A, B, C, D, E, F) with the following FDs:
AB → C, C → D, D → E, AE → F

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes:
D1: R = [(A, B, C); (C, D); (D, E); (A, E, F)]
D2: R = [(A, B, C, D); (D, E); (A, E, F)]

(a) Check whether D1 and D2 are Lossy or Lossless Join Decompositions. Also determine all the candidate keys of R.

(b) Consider a relation R (A, B, C, D, E) with FDs: AB → C, C → D, D → E, and E → A.
i. Determine all the candidate keys of relation R.
ii. Determine the normal form of R (check up to BCNF). Show all steps of decomposition up to BCNF, giving proper justification at each step.</td><td>20</td></tr>
<tr><td>5</td><td>(a) Explain the different types of triggers in DBMS. Differentiate between BEFORE and AFTER triggers with examples.

(b) Write a PL/SQL function that accepts an integer as input and returns 'Prime' if the number is prime and 'Not Prime' otherwise. Demonstrate the function using a PL/SQL block</td><td>20</td></tr>
<tr><td>6</td><td>(a) Consider the following schedule involving three transactions T1, T2, and T3. Determine whether it is conflict serializable. If serializable, identify the equivalent serial schedule.
<table>
<tr><th>T1</th><th>T2</th><th>T3</th></tr>
<tr><td>READ (A)</td><td></td><td></td></tr>
<tr><td></td><td>READ (B)</td><td></td></tr>
<tr><td>WRITE (A)</td><td></td><td></td></tr>
<tr><td></td><td></td><td>READ (A)</td></tr>
<tr><td></td><td>WRITE (B)</td><td></td></tr>
<tr><td></td><td></td><td>WRITE (A)</td></tr>
</table>
(b) Consider a banking transaction in which ₹10,000 is transferred from Account A to Account B. Explain how the ACID properties ensure the correctness and reliability of this transaction.</td><td>20</td></tr>
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**** You Can See Your Sheets on 9th Sept, 2026 at 3:30 in L557, CSED Building.**